

Heart Disease Prediction Using Machine Learning

1. Abstract

Heart disease is one of the leading causes of death worldwide. This project uses machine learning techniques to predict whether a person is likely to have heart disease based on various health indicators. The project includes data preprocessing, feature engineering, exploratory data analysis, model training, evaluation, and prediction.

2. Problem Statement

Heart disease often remains undetected until severe symptoms appear. Therefore, an intelligent prediction system is needed to assist healthcare professionals by providing quick and accurate predictions based on patient health data.

3. Objectives

- Predict whether a person has heart disease.
- Perform data preprocessing and cleaning.
- Create meaningful features from existing data.
- Train multiple Machine Learning models.
- Compare model performances.
- Select the best model for prediction.

4. Dataset Description

The project uses the Heart Disease Health Indicators dataset containing features such as Age, BMI, High Blood Pressure, High Cholesterol, Diabetes, Smoking, Physical Activity, Fruits, Vegetables, Alcohol Consumption, General Health, Mental Health, Physical Health, Education and Income. The target variable is HeartDiseaseorAttack (0 = No, 1 = Yes).

5. Software Requirements

- Programming Language : Python
- Platform: Google Colab
- Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib

6. Methodology

Step 1: Import Libraries

The required Python libraries are imported for data manipulation, visualization, machine learning, and model saving.

Step 2: Load Dataset

The dataset is loaded into a Pandas DataFrame.

Step 3: Data Preprocessing

The following preprocessing steps are performed:

- Checked dataset information
- Displayed statistical summary
- Checked missing values
- Removed unnecessary columns:
 - CholCheck
 - AnyHealthcare

Step 4: Feature Engineering

Three new features were created to improve prediction accuracy.

1.Health Score

Combines

- General Health
- Physical Health
- Mental Health

Formula:

Health Score = GenHlth + PhysHlth + MentHlth

2.Lifestyle Score

Measures healthy lifestyle habits.

3.Metabolic Risk Score

Measures metabolic risk using

- High Blood Pressure ,diabetes
- High Cholesterol ,IBM

7. Exploratory Data Analysis (EDA)

Two visualizations were created.

Graph 1

Heart Disease Distribution

Purpose:Shows the number of patients with and without heart disease.

Graph 2

Age vs BMI Scatter Plot

Purpose: Shows the relationship between Age and BMI while distinguishing heart disease

8. Data Splitting

The dataset is divided into

- 80% Training Data
- 20% Testing Data

Stratified sampling is used to maintain equal class distribution.

9. Machine Learning Models Used

1. Logistic Regression

Purpose:

A simple classification algorithm that predicts probabilities using a logistic function.

Advantages

- Fast
 - Easy to understand
 - Works well for binary classification
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2. K-Nearest Neighbors (KNN)

Purpose:

Predicts the class based on the nearest neighboring data points.

Advantages

- Simple
 - No training assumptions
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3. AdaBoost Classifier

Purpose:

Combines multiple weak learners into one strong learner.

Advantages

- Improves prediction accuracy
- Reduces bias

4. Gradient Boosting Classifier

Purpose:

Builds multiple decision trees sequentially, where each tree corrects the errors of the previous one.

Advantages

- High prediction accuracy
- Handles complex datasets effectively
- Performs well on structured/tabular healthcare data

10. Model Evaluation

Each model is evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

These metrics help determine the best-performing algorithm.

11. Prediction Process

The trained Gradient Boosting model predicts whether a new patient has heart disease.

Output

- 0 → No Heart Disease
- 1 → Heart Disease

12. Future Scope

Develop a web application using Flask or Streamlit, deploy the model on the cloud, integrate Explainable AI, enable real-time patient data input, and connect with hospital databases.

13. Conclusion

This project successfully demonstrates the use of machine learning for heart disease prediction. After preprocessing, feature engineering and comparison of different algorithms, the Gradient Boosting Classifier was selected as the final model because of its superior performance.